Jaydyn Davis

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Professional Summary

Hardware reverse engineer and vulnerability researcher with an Interim Secret clearance and Battelle Mission Focused Tools experience across embedded ARM/MIPS/RISC targets, firmware debugging, Linux kernel development, and offensive cyber R&D.

Education

Hofstra University, Hempstead, NY
Bachelor of Science in Computer Science & Cybersecurity (Expected December 2027) | GPA: 3.30

Honors & Activities

2nd Place, Hofstra-Amazon HUCTF - $4,000 team prize from a $12,000 prize pool
Dean's List (5x) | National Society of Leadership and Success | Upsilon Pi Epsilon Computer Science Honor Society

Relevant Coursework

Operating Systems, Programming Languages, Data Structures & Algorithms, Network Security, Ethical Hacking, Data Communications & Networking, Cryptography

Work Experience

Battelle - *Reverse Engineering / Vulnerability Research Co-op, Mission Focused Tools (MFT)*
Columbus, OH | August 2026 - December 2026

- Reverse engineered embedded ARM, MIPS, and other RISC-based targets using Ghidra, Binary Ninja, GDB/pwndbg, C, and assembly; traced control flow, registers, memory state, and system behavior.
- Implemented a firmware memory debugger and C Linux kernel module providing controlled device-memory read/write access for authorized vulnerability research.
- Built a research automation pipeline integrating web-scraped inputs with system/tool logs; configured Linux services, path-triggered tasks, and timers and packaged dependencies with Docker.
- Combined network attack analysis with low-level vulnerability research in authorized offensive cyber R&D workflows; documented findings and collaborated with multidisciplinary engineers.

The Coder School - *Coding Coach*
Bellmore, NY | March 2023 - November 2023

- Taught project-based Python and Scratch to 100+ students, strengthening technical communication, debugging, and problem decomposition.

Reverse Engineering & Security Research

Binary Analysis & Exploit Development | Authorized Academic Labs

- Analyzed x86/x64 Linux and Windows binaries with GDB, Ghidra, and Immunity Debugger, mapping C/C++ to assembly and tracing instructions, registers, stack frames, system calls, and control flow.
- Reproduced controlled stack/heap overflows, format-string vulnerabilities, shellcode behavior, command injection, and remote exploitation; calculated offsets and evaluated exploit-mitigation concepts.
- Developed and debugged custom Metasploit modules and Ruby/Perl exploit workflows; documented reproducible evidence, root cause, technical impact, and defensive recommendations.

SecPipe & Rodman | Python Security Research Automation

- Built a modular Python pipeline that normalizes Linux authentication, Nginx, syslog, and JSON events and applies ATT&CK-mapped detections through extensible parser, detection, and output registries.
- Developed a Python analysis pipeline combining AI-assisted classification, reputation services, and disposable Multipass VM sandboxes to evaluate suspicious URLs, network behavior, and TLS metadata.
- Used Git, pytest, GitHub Actions, Docker, structured logging, and technical documentation to support repeatable testing, integration, and deployment.

Technical Skills

- Languages/Architectures: C, C++, Python, Bash/Shell, Ruby, Perl, C#, x86/x64 Assembly, ARM, MIPS, RISC, computer architecture
- Reverse Engineering/Debugging: Ghidra, Binary Ninja, GDB/pwndbg, radare2, Immunity Debugger, IDA Free familiarity, static/dynamic analysis, disassembly/decompilation
- Embedded/Kernel: firmware analysis, debugger development, Linux kernel modules, controlled device-memory investigation, operating-system internals
- Vulnerability Research: memory corruption, stack/heap analysis, shellcode, format strings, command injection, proof-of-concept development, exploit mitigations
- Systems/Network: Linux administration, services, path-triggered tasks, timers, Docker, Git, Wireshark, tcpdump, Nmap, Netcat, Burp Suite, Metasploit